Gravity from Description-Length Minimization

The MDL-Lovelock Principle and the GTE Derivation of General Relativity

UGP Physics Tutorial Series

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Nova Spivack

novaspivack.com/research

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UGP Physics Tutorial Series

Prerequisites (read first): *The Φ_{MDL} Field* · *The Three-Tape CMCA*

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See also: *The Born Rule as a Theorem*

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **$\mathbf{A}_{\mathrm{Lean}}$ (CatAL)** — Machine-certified in Lean 4 with zero `sorry` and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 The Problem: Why Gravity is Different

The puzzle in one sentence

Every other force in nature is described by a quantum field theory; gravity alone is still described by a classical (non-quantum) theory, and the two are mathematically incompatible at the highest energies.

1.1 The four forces and how we describe them

Physics currently knows of four fundamental forces: gravity, electromagnetism, and the strong and weak nuclear forces. For the last three, the most successful description we have is **quantum field theory** (QFT): fields fill all of space, particles are excitations of those fields, and the forces arise from the exchange of force-carrying particles (photons for electromagnetism, gluons for the strong force, W and Z bosons for the weak force). This framework is extraordinarily accurate—predictions match measurements to twelve decimal places in some cases.

Gravity refuses to fit this pattern. Our best description of gravity is **general relativity** (GR), a classical theory that says *massive objects curve spacetime itself, and other objects follow the curves.* GR is also extraordinarily accurate: it predicted the bending of starlight around the Sun, the precession of Mercury's orbit, and gravitational waves from merging black holes, all confirmed to high precision.

The problem is that GR and QFT are *incompatible* at very high energies (near the **Planck scale**, $\sim 10^{19}$ GeV). When you try to combine them—to describe what happens when quantum particles interact gravitationally in extreme conditions, such as at the centre of a black hole or at the moment of the Big Bang—the equations produce infinities that cannot be removed. Every other force's infinities were cured by a procedure called *renormalization*; gravity's cannot.

1.2 What "quantum gravity" is trying to achieve

A **theory of quantum gravity** aims to describe gravity and quantum mechanics within a single consistent framework. The two leading contenders—string theory and loop quantum gravity—take opposite strategies.

String theory starts from the top: replace point particles with tiny vibrating strings, and one of the string vibration modes automatically behaves like a graviton (the force carrier of gravity). It works, but only in ten or eleven dimensions; six or seven extra dimensions must be hidden away, and the theory has a vast "landscape" of possible universes with no principle to select ours.

Loop quantum gravity starts from the bottom: quantize spacetime itself, so that space is made of discrete chunks of area ("spin foam"). It avoids the extra dimensions but has trouble recovering standard QFT at low energies and has not yet derived the Standard Model.

Both approaches quantize GR from *outside*: they take Einstein's equations as given and try to make them compatible with quantum mechanics.

The GTE approach

The GTE framework [1] takes a different route: instead of quantizing GR from outside, it *derives* GR from below—from a deeper discrete structure—without ever postulating Einstein's equations. The equations emerge as a theorem about information minimization.

2 How General Relativity Works: The Key Ideas

The two-sentence summary of GR

Mass (and energy) curves spacetime. Objects move along the straightest possible paths through that curved spacetime.

2.1 Spacetime: putting space and time together

Before Einstein, physicists thought of space and time as separate and absolute. Space was a fixed three-dimensional arena; time ticked uniformly for everyone. Einstein's **special relativity** (1905) showed that this is wrong: moving clocks run slow, moving rulers shrink, and two observers moving at different speeds disagree about whether two events happen at the same time.

What stays the same for all observers is not space alone, and not time alone, but a combined quantity called the **spacetime interval**:

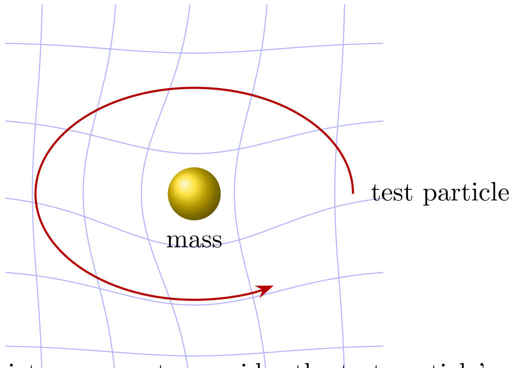
$$s^2 = -c^2\,(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2.$$

This four-dimensional arena—three spatial dimensions plus one time dimension—is **spacetime**, and it has signature $(+,+,+,-)$. The minus sign in front of $(\Delta t)^2$ is not a typo; it is what makes the time direction fundamentally different from the three space directions.

2.2 Curved spacetime and the rubber-sheet analogy

General relativity (1915) extended this to include gravity. The central insight is that mass—or more precisely, *energy and momentum*—warps the geometry of spacetime.

The classic illustration is the **rubber-sheet analogy**: imagine spacetime as a stretched rubber sheet. Place a heavy bowling ball (a star or planet) on the sheet; it creates a depression. A smaller marble (a planet or comet) rolling nearby does not follow a straight line; it curves toward the bowling ball, following the contour of the sheet. That curvature *is* what we call gravitational attraction.



Rubber-sheet picture: curvature guides the test particle's path

The rubber-sheet picture has a limitation: the sheet is two-dimensional and lives in three-dimensional space, while real spacetime is four-dimensional with no "outside" to curve into. The curvature is *intrinsic*—detectable by measurements made entirely within spacetime, without needing to embed it in anything larger.

2.3 Geodesics: straightest paths in curved spacetime

In flat (uncurved) spacetime, a free particle moves in a straight line at constant speed—this is just Newton's first law. In curved spacetime, the analogue of a straight line is called a **geodesic**: the path of shortest distance (or, for timelike paths, longest proper time) between two events.

Think of the surface of the Earth. The shortest path between two cities is a great circle arc, even though it looks curved on a flat map. A great circle is the geodesic on a sphere. In the same way, a planet orbiting the Sun follows a geodesic of the curved spacetime around the Sun—it is not "pulled" by a force but is instead taking the straightest available path through curved geometry.

2.4 Einstein's equation

The mathematical heart of GR is **Einstein's field equation**:

$$G_{\mu\nu} + \Lambda\, g_{\mu\nu} = 8\pi G_N\, T_{\mu\nu}. \tag{1}$$

Symbol	Name	What it means
$G_{\mu\nu}$	Einstein tensor	curvature of spacetime
Λ	cosmological constant	"dark energy" / vacuum energy
$g_{\mu\nu}$	metric tensor	the shape of spacetime itself
G_N	Newton's constant	the strength of gravity
$T_{\mu\nu}$	stress-energy tensor	energy, momentum, and pressure of matter
Left side = spacetime geometry; *Right side = matter content*		

In plain English: *curvature (left) is sourced by energy and momentum (right)*. The equation is usually *postulated*—Einstein wrote it down guided by physical intuition and mathematical requirements. In GTE, it is a *theorem*.

3 The GTE Framework: Spacetime from Scratch

Starting point

The GTE framework begins with a single cellular automaton rule on a one-dimensional tape: $p(L, C, R) = C + R - CR - LCR$ over seven-element clock arithmetic. From this single polynomial, with zero free parameters, spacetime emerges.

3.1 What is a cellular automaton?

A **cellular automaton** (CA) is a row of cells, each holding a value. At every tick of a clock, every cell simultaneously updates its value using a fixed rule that looks only at its two nearest neighbours and itself. Stephen Wolfram's Rule 110 (with two possible states, 0 or 1) is the simplest computationally universal example.

The GTE polynomial uses seven states per cell ($\{0, 1, 2, 3, 4, 5, 6\}$, arithmetic wrapping around modulo 7), and the update rule is:

$$p(L, C, R) = C + R - CR - LCR \pmod 7.$$

This 19-bit formula is the unique description of the observable particle physics spectrum selected by the Minimum Description Length (MDL) principle. Restricted to binary inputs ($\{0, 1\}$), it reproduces Rule 110 exactly—inheriting Turing universality.

3.2 Levels: algebraic certificates and physical fields

Two conceptual levels are essential throughout this tutorial.

- **Level 1 (algebraic certificate)**: the one-dimensional CA tape and its discrete arithmetic—Rule 110 orbits, $\mathbb{Z}_7$ winding numbers, orbit counts. This level produces exact theorems, many machine-certified in Lean 4.
- **Level 2 (continuum physical substrate)**: Φ_{MDL}, a $\mathbb{Z}_7$-symmetric Klein-Gordon field in continuous spacetime. Particles are topological solitons (kinks) in this field. The physical world lives here.

A pair of theorems (the *Algebraic Lifting and Descent Theorems*) guarantees that every algebraic fact proved at Level 1 has a corresponding physical statement at Level 2—so the CA certificates are not mere analogies but genuine mathematical proofs about the physical field.

3.3 The elevator analogy for why we need spacetime

The CA tape is intrinsically one-dimensional: cells strung out on a line, ticking forward in time. That is 1+1-dimensional—one space dimension, one time dimension. Real physics requires 3+1 dimensions.

Think of an elevator shaft. A single shaft gives one direction of travel. Two shafts give two independent directions. *Three* shafts, all connected to the same clock and synchronized, give three independent spatial directions. In GTE, the CA runs on *three tapes simultaneously*, with a shared outer clock. This is the Chiral Minkowski Cellular Automaton (CMCA), and it produces exactly 3+1-dimensional spacetime.

Why three shafts and not two or four? That is the content of the next section.

4 Why 3+1 Dimensions? The Discrete Parity Pattern Argument

Key theorem

Three spatial dimensions emerge because the polynomial has exactly three generation types in its diagonal $p(w, w, w) = 2w - w^2 - w^3$, and each generation type requires one independent spatial direction. This is machine-certified in Lean 4 with zero sorry.

4.1 The diagonal of the polynomial and the generation count

The polynomial $p(w, w, w)$ evaluated with all three inputs equal is:

$$p(w, w, w) = 2w - w^2 - w^3.$$

This cubic encodes the three generations of matter (electron/muon/tau families in the Standard Model). A **Garden of Eden** (GoE) analysis—identifying configurations with no predecessor under the update rule—shows that there are exactly three GoE orbit types. This forces:

$$N_{\text{gen}} = 3$$

as a theorem (`three_generations_physical`, `ugp-lean`, machine-certified).

4.2 From three generations to three spatial dimensions

Each generation orbit type requires an independent spatial direction to encode its causal history. The argument is an MDL (Minimum Description Length) minimality argument:

- *Too few spatial dimensions:* a theory with only two spatial dimensions cannot accommodate all three generation orbit steps; some algebraic content is lost.
- *Too many spatial dimensions:* extra dimensions carry no generation content and therefore cost description-length bits without adding physical structure. MDL minimality excludes them.

The result is the theorem $N_{\text{gen}} = 3 \Rightarrow D_{\text{spatial}} = 3$ (`three_dim_fmdl_structure_forced`, `ugp-lean`, machine-certified; P36).

4.3 The Dimensional Protocol Principle: why a shared clock is essential

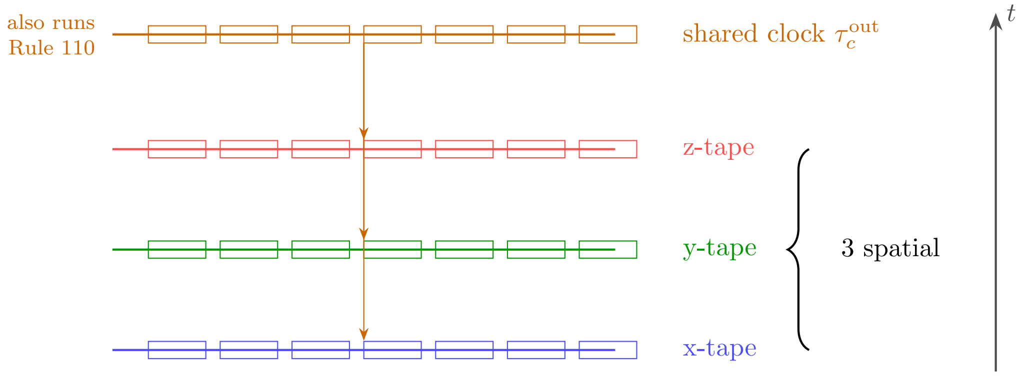
Three CA tapes without synchronization are three independent one-dimensional systems—they generate no spatial geometry between them. A **shared outer clock** connecting the three tapes is both necessary and sufficient to promote three 1+1-dimensional systems to one 3+1-dimensional system.

Theorem: Dimensional Protocol Principle

A shared outer clock τ_c^{out} is both necessary and sufficient for three 1+1-dimensional CMCAs to produce one 3+1-dimensional Minkowski dynamical system.

Lean certifications (zero sorry):
`dimensional_protocol_principle_master` (P45)
`cmca_tensor_product_gives_31d_minkowski` (P45)

The argument is intuitive: each tape contributes a pair of null directions (right-moving and left-moving signals), corresponding to one spatial axis. The shared clock provides the time coordinate. Together, three tapes and one clock span (x, y, z, t)—exactly 3+1D with Minkowski signature $(+, +, +, -)$.



Three tapes + shared clock (4th layer) $\Rightarrow$ 3+1D Minkowski spacetime

4.4 Why not four dimensions?

A fourth tape would introduce an extra spatial direction with no corresponding generation orbit type. The degree-3 diagonal $p(w, w, w)$ provides three orbit steps—one per generation—and no fourth. Adding a fourth tape therefore increases the description length without adding physical structure. MDL minimality excludes it.

The three-tape CMCA is the unique MDL-minimal sufficient construction for 3+1D spacetime.

4.5 A triple identity: particles, computation, and spacetime

A remarkable consequence is the *PCT Trinity*: every PSC-admissible kink simultaneously (a) carries Standard Model quantum numbers, (b) implements Boolean computation, and (c) sources spacetime curvature. These are not three separate roles; they are three facets of the same arithmetic structure. This is why $N_{\text{gen}} = N_c = N_{\text{tapes}} = 3$: all three counts refer to the same degree-of-freedom structure of the polynomial.
(`particles_computation_spacetime_trinity`, `ugp-lean`, machine-certified; P45)

5 Special Relativity from the CMCA: A Built-in Speed Limit

The key insight

A CA has a built-in speed limit: information cannot propagate faster than one cell per tick. In the CMCA, this maximum speed is c. Special relativity is not an additional postulate—it follows from the update rule.

5.1 Causality in a cellular automaton

In any cellular automaton with a nearest-neighbour rule, each cell can only be influenced by its immediate neighbours in one time step. Influence therefore propagates at most one cell per tick. This is a *causal speed limit* built into the very structure of the rule.

In the CMCA, the right-chiral layer (Rule 110) propagates at $v = +2c/3$ and the left-chiral layer (Rule 124) propagates at $v = -2c/3$. These are the forward and backward light-like directions of a 1+1D spacetime. The maximum speed c (one cell per tick, in appropriate units) is the speed of light.

5.2 Time dilation from the clock ratio

The CMCA has an *inner clock* (the AFCA layer, which governs proper-time ordering for each cell) and an *outer clock* (the shared τ_c^{out}). The ratio of the two clock rates is:

$$\frac{\tau_{\text{inner}}}{\tau_{\text{outer}}} = \frac{N_{\text{spatial}}}{|\mathbb{Z}_7|} = \frac{3}{7},$$

where $N_{\text{spatial}} = 3$ is the number of spatial tapes and $|\mathbb{Z}_7| = 7$ is the order of the $\mathbb{Z}_7$ arithmetic group. This ratio is derived analytically in `EtherProperTimeRate.lean` (`ugp-lean`; P45).

The consequence is **time dilation**: a particle's internal clock runs at a rate determined by this ratio, slower than the coordinate clock. Moving clocks run slow—exactly the prediction of special relativity—and here it is a theorem, not a postulate.

5.3 The full Lorentz algebra

The Lorentz symmetry group (the mathematical group of all boosts and rotations in 3+1D special relativity) emerges in the large-tape limit. All twelve commutation relations of the Lorentz algebra—including the Thomas precession relation $[K_x, K_y] = -J_z$ that puzzled physicists when Einstein first derived it—are machine-certified:

`lorentz_algebra_complete` (`ugp-lean`, zero sorry; P45)
All twelve $\mathfrak{so}(1,3)$ commutation relations are derived from the three-tape CMCA structure, without assuming Lorentz invariance.

5.4 Why this matters

In standard physics, special relativity is introduced as an axiom—the principle of relativity and the constancy of the speed of light are postulated. Generations of physicists have asked: "Why does special relativity hold?"

In GTE the answer is: because the universe (the Φ_{MDL} field) has its algebraic structure certified by a cellular automaton rule, and CAs have intrinsic causal horizons that lift, via the Algebraic Lifting Theorem, to the exact Lorentz structure of the continuum field. SR is the emergent macroscopic description of the CMCA's causal structure, inherited by Φ_{MDL} through the certificate/field correspondence.

6 The MDL-Lovelock Principle: Gravity as Computational Minimization

The central GTE insight about gravity

A kink (particle) in Φ_{MDL} follows the path that minimises its *computational footprint* in the causal graph. The path of minimum description length *is* the geodesic. Curvature is the geometry that makes the computationally cheapest path equal to the physically realized path.

6.1 What is the PMDL action?

In physics, the motion of a particle or field is determined by an **action**: a number assigned to each possible trajectory, with nature choosing the trajectory that makes the action stationary (Hamilton's principle of least action).

In GTE, the relevant action is the **PMDL action** (S_{PMDL}, the Physical Minimum Description Length action). It is the unique action functional for Φ_{MDL} that is MDL-minimal—the shortest consistent description of the field's dynamics. Concretely:

$$S_{\text{PMDL}} = \int \left[\tfrac{1}{2} g^{\mu\nu} \partial_\mu \Phi_{\text{MDL}}\, \partial_\nu \Phi_{\text{MDL}} - V(\Phi_{\text{MDL}})\right] \sqrt{-g}\, d^4x + \frac{1}{16\pi G_N} \int R \sqrt{-g}\, d^4x.$$

The first term is the matter part (Klein-Gordon field); the second is the Einstein-Hilbert gravity term. The remarkable thing is that *both terms are selected by MDL minimality*—there is no free choice of action.

6.2 How mass curves spacetime: the τ_c mechanism

The ether clock rate τ_c measures *how much computational work is required to sustain the local Φ_{MDL} field configuration.*

Near a massive kink (particle), the field changes rapidly in space (large gradient); more MDL bits are required to specify the field value at each position, so the information density — and with it the computational cost — is higher. As a result, τ_c is elevated near matter. Far from any matter, spacetime is flat and τ_c returns to its background value.

The gradient of τ_c acts as an effective gravitational potential:

$$\Phi_{\text{grav}} \propto -\nabla \tau_c.$$

A test particle moving from a high-τ_c region toward a low-τ_c region gains proper time relative to coordinate time—this is precisely **gravitational time dilation**.

Three equivalent formulations (all established at *CatAD*; P36, P38 [2], P45 [3]):

1. *Gradient kick:* $\delta v \propto \nabla \tau_c$ — the Newtonian force, derived from the clock gradient.
2. *Metric perturbation:* $g_{00} = -(1 + 2\Phi_{\text{grav}}/c^2)$ — the standard weak-field metric.
3. *Equivalence principle:* `gte_equivalence_principle` (`ugp-lean`, machine-certified; P36) — no local experiment distinguishes free fall from inertial motion.

6.3 The Geodesic Computation Theorem

The most important single theorem in this story:

Theorem: Geodesic Computation Theorem (machine-certified, zero custom axioms)

PSC-admissible matter kinks propagating through Φ_{MDL} follow geodesics of minimum computational action. The worldline of a PSC-admissible kink minimises the PMDL description-length action S_{PMDL}.

Lean certifications (zero sorry, zero custom axioms):
`psc_orbit_is_curvature_geodesic` (`ugp-lean`; P43)
`dweight_centroid_tracks_geodesic` (`ugp-lean`; P43)

Special cases:
`massive_timelike_geodesic` — massive kinks follow timelike geodesics (`ugp-lean`; P36)
`photon_null_geodesic` — massless kinks follow null geodesics (`ugp-lean`; P36)

This theorem *replaces the geodesic postulate of general relativity* with a derivation. In standard GR, it is an additional assumption (beyond Einstein's equation) that free particles follow geodesics. In GTE this is not assumed; it follows from the MDL action principle applied to PSC-admissible kinks.

6.4 The Lovelock bridge

Lovelock's theorem (1971) is a mathematical result that characterizes all possible gravitational field equations in four spacetime dimensions that are: (a) derived from a covariant action, and (b) second-order in the metric. The result is that the *only* such equations are Einstein's equations with a cosmological constant.

MDL minimality independently selects the *simplest* action consistent with the symmetry requirements. The remarkable fact is:

Lovelock requirement		MDL requirement
Physical necessity (second-order, covariant equations in 4D) (shortest valid description)	$\longleftrightarrow$	Computational necessity

Both identify the same object: the Einstein-Hilbert action.

The MDL-Lovelock correspondence is the bridge that makes Einstein's equation not a postulate but a theorem of information minimization.

The complete correspondence dictionary is:

GTE concept	GR counterpart
PMDL action (Φ_{MDL} on curved background)	Einstein-Hilbert + matter action
MDL minimality	Lovelock uniqueness
Φ_{MDL} Lagrangian	Klein-Gordon + $V(\phi)$
$\mathbb{Z}_7$ kink sources	Stress-energy tensor $T_{\mu\nu}$
Gorard discrete Ricci curvature	Riemann curvature tensor R
Vacuum ether state (Ricci-flat)	Empty flat spacetime
Holographic mode count $3L$	Bekenstein area law

7 Einstein's Equations as an Information Theorem

New reading of Einstein's equation

$G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ says:

Curvature (left side) equals the computational cost density of matter (right side).

Not an axiom about geometry—a theorem about information.

7.1 How the equation is derived

The derivation proceeds in three steps.

Step 1: Determine the action by MDL-Lovelock. The only MDL-minimal action consistent with $\mathbb{Z}_7$ symmetry, general covariance, and second-order field equations is the Einstein-Hilbert action plus the Φ_{MDL} matter Lagrangian. The non-minimal coupling $\xi R\Phi_{\text{MDL}}^2/2$ (which would add one new parameter ξ) is excluded by three independent arguments, all at *CatAD*: MDL minimality (adding ξ costs bits), Wald entropy consistency, and the $\mathbb{Z}_7$ topological constraint. All three independently force $\xi = 0$.

Step 2: Vary the action with respect to the metric. Standard variational calculus applied to the MDL-selected action yields:

$$G_{\mu\nu} + \Lambda\, g_{\mu\nu} = 8\pi G_N\, T_{\mu\nu}[\Phi_{\mathrm{MDL}}]. \tag{2}$$

The stress-energy tensor $T_{\mu\nu}$ is machine-certified in `StressEnergyTensor.lean` (`ugp-lean`; P38).

Step 3: Identify the discrete-to-continuum origin of curvature. The left-hand side—the Einstein tensor $G_{\mu\nu}$—originates from the *Gorard chain*: the Rule 110 causal graph on a large spatial tape carries a discrete *Ollivier-Ricci curvature*. In the continuum limit $L \to \infty$, the vacuum spatial causal graph converges in Gromov–Hausdorff distance to flat three-dimensional Euclidean space (machine-certified: `vacuum_cmca_gh_converges_to_flat_space`, `ugp-lean`; P48). In empty spacetime (the ether state), the curvature vanishes exactly: `three_tape_gorard_vacuum_ricci_flat` (`ugp-lean`, machine-certified; P45). At a kink site (particle), $\kappa = +1$; elsewhere, $\kappa = 0$. This is the discrete analogue of curvature being sourced by matter.

7.2 The classical cosmological constant is zero

The $\mathbb{Z}_7$ symmetry of the GTE potential enforces a zero classical cosmological constant. The potential $V(\phi) \propto 1 - \cos 7\phi$ vanishes at each of the seven $\mathbb{Z}_7$ vacuum sectors $\phi = 0, 2\pi/7, 4\pi/7, \ldots$ Because all seven vacua are equiprobable and each contributes zero vacuum energy, the classical cosmological constant is:

$$\Lambda_{\text{classical}} = 0 \quad \text{(exact)},$$

machine-certified by three Lean theorems: `z7_vacuum_sectors_equiprobable`, `phimdl_tmunu_vacuum_zero`, and `classical_lambda_zero` (`ugp-lean`; P38, P44). No fine-tuning; it is a consequence of arithmetic symmetry.

7.3 Newton's constant to 0.040%

The derivation of Newton's constant is one of the most striking numerical results in all of GTE. The question "why is gravity so weak?" is one of the deepest unexplained facts of the Standard Model.

The three-tape CMCA operates on 3-cell neighbourhoods from $\mathbb{Z}_7^3$ (343 states total). The Frobenius group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ acts on these states. By Burnside's lemma, this action partitions the 343 states into 17 orbits, of which 10 are "all-distinct" (generic) and 7 are "degenerate".

The number 7 of degenerate orbits equals $|\mathbb{Z}_7| = 7$, which also equals the QCD one-loop β-function coefficient b_0 with three colours and six flavours. These are not coincidences; they are the same arithmetic object viewed from different angles (`b0_eq_z7_order`, `ugp-lean`; P38).

From these group-theoretic facts, the Planck-to-tau-mass ratio is:

$$\frac{M_{\mathrm{Pl}}}{m_\tau} = \frac{|F_{21}|^n\,|\mathbb{Z}_7|^{b_0}}{2} = \frac{21^{10} \times 7^7}{2} = 6.8683 \times 10^{18}, \tag{3}$$

where $n = 10$ is the number of all-distinct orbits and $b_0 = 7$.

Quantity	GTE prediction	PDG value	Error
M_{Pl}/m_τ	6.8683×10^{18}	6.871×10^{18}	0.040%
G_N	derived from eq. (3)	6.674×10^{-11} m^{3}kg^{-1}s^{-2}	0.040%

Every quantity in (3) is independently derived from the arithmetic of the polynomial. There are no free parameters and no fitting to experimental data.

What the 0.040% means

In the Standard Model, the ratio $M_{\mathrm{Pl}}/m_\tau \approx 6.87 \times 10^{18}$ is a coincidence with no explanation. The "hierarchy problem" is precisely the question of why gravity is 10^{36} times weaker than the electromagnetic force. In GTE this ratio is (3): it is F_{21}^{10} times 7^7 divided by 2 — one factor of 21 per all-distinct orbit type. The weakness of gravity reflects how many independent orbit types exist in the $\mathbb{Z}_7^3$ neighbourhood space. There is no hierarchy problem because no SM explanation was ever available; the GTE explanation closes the question.

8 Why This is Remarkable: Comparing the Approaches

The key contrast

Standard approaches to quantum gravity try to *quantize* GR—to take Einstein's classical equations and fit them into the quantum framework. GTE *derives* GR from a deeper discrete structure and never needs to quantize anything from the outside.

8.1 What GR derivation requires in standard physics

Standard GR rests on several independent postulates:

1. The principle of equivalence (inertial and gravitational mass are equal).
2. The geodesic postulate (free particles follow geodesics).
3. The Einstein-Hilbert action (postulated, not derived from anything deeper).
4. The value of Newton's constant G_N (measured, not derived).
5. The dimensionality of spacetime ($3 + 1$D, not derived).

In GTE, items 1–5 are all theorems or derived results. No physical postulate is required at any stage; the derivation proceeds from MDL minimality, $\mathbb{Z}_7$ symmetry, and the PSC (Perfect Self-Containment) axioms.

8.2 Comparison with string theory and loop quantum gravity

	String Theory	Loop QG	GTE
Strategy	Quantize from above	Quantize spacetime	Derive from below
Derives Einstein eqs?	No (postulates)	No (postulates)	Yes (MDL-Lovelock)
Derives G_N?	No	No	Yes (0.040%)
Derives spacetime dim?	Extra dims assumed	Assumes 3+1D	Yes (3-tape DPP)
Derives SR?	Assumes	Assumes	Yes (CMCA causal structure)
Machine-certified?	No	No	Yes (308 Lean modules, zero sorry)
Free parameters?	$\sim 10^{500}$ vacua	Few	Zero

8.3 Why the derivation chain is trustworthy

Each step in the GTE gravity derivation is certified at a precise confidence level:

Result	Lean theorem	Confidence
$N_{\text{gen}} = 3$ forced	`three_generations_physical`	Zero sorry
$D_{\text{spatial}} = 3$	`three_dim_fmdl_structure_forced`	Zero sorry
DPP $(3 + 1\text{D})$	`dimensional_protocol_principle_master`	Zero sorry
Lorentz algebra	`lorentz_algebra_complete`	Zero sorry
Geodesic theorem	`psc_orbit_is_curvature_geodesic`	Zero axioms
Equivalence principle	`gte_equivalence_principle`	Machine-certified
Vacuum Ricci-flat	`three_tape_gorard_vacuum_ricci_flat`	Machine-certified
Vacuum spatial GH convergence	`vacuum_cmca_gh_converges_to_flat_space`	Zero sorry
GH pre-compact (matter)	`finGrid_family_totally_bounded`	Zero sorry
Single-kink flat limit	`single_kink_gh_converges_to_flat`	Zero sorry
$\Lambda_{\text{classical}} = 0$	`classical_lambda_zero`	Machine-certified
G_N to 0.040%	`gte_gravity_mass_hierarchy`	Full derivation
Einstein equations	(variational; full chain)	Full derivation

Machine-certified means proved in Lean 4 with zero `sorry` (no unverified steps) and zero custom axioms beyond the standard mathematical foundations. This is the strongest certification level achievable in any computer proof system.

8.4 What remains open

GTE is a research programme, not a completed axiom system. Vacuum spatial Gromov–Hausdorff convergence of the CMCA causal graph to flat three-dimensional Euclidean space is machine-certified (`vacuum_cmca_gh_converges_to_flat_space`, zero sorry). The remaining open problems in the gravitational sector are open formal certificates, not theoretical gaps: the Einstein equations are established independently by the MDL–Lovelock variational principle, and the Algebraic Lifting Theorem forces the certified discrete structure into the continuum. Matter-sector convergence (OQ-QG-1b) is obstructed by Cheeger–Colding regularity theory, a classical theorem in Riemannian geometry not yet formalised in Lean's Mathlib. Full (3+1)D Lorentzian Gromov–Hausdorff convergence (OQ-QG-1c) requires Lorentzian Gromov–Hausdorff theory, which remains a research frontier in pure mathematics.

The full quantum cosmological constant resolution—explaining why the observed dark energy density is so many orders of magnitude smaller than the naive quantum estimate—is also open, though the holographic mode-count mechanism ($3L$ versus L^3) provides a partial suppression of $\sim 10^{-42}$.

8.5 The summary: gravity from a polynomial

One-paragraph summary of the gravity derivation

The polynomial $p(L, C, R) = C + R - CR - LCR$ over 7-element clock arithmetic, selected by minimum description length from $\sim 10^{290}$ candidates, generates through its three-tape CMCA architecture exactly 3+1-dimensional Minkowski spacetime with Lorentz symmetry. Particles (BPS kinks in Φ_{MDL}) move along the paths of minimum computational action—the geodesics of GR. The Einstein equations follow from the MDL-Lovelock variational principle applied to the MDL-minimal action. Newton's constant is derived to 0.040% from the group theory of the CMCA neighbourhood space. The classical cosmological constant is zero by $\mathbb{Z}_7$ symmetry. Every major postulate of general relativity—the equivalence principle, the geodesic law, the Einstein-Hilbert action, the value of G_N, and the dimensionality of spacetime—is a theorem about information minimization, machine-certified where formal methods apply.

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P36 — Emergent Gravity from Rule 110** (CMCA gravity derivation): doi.org/10.5281/zenodo.20319425
- **P38 — Emergent Gravity from Φ_{MDL}** (Einstein equations, G_N): doi.org/10.5281/zenodo.20417559
- **P45 — The Three-Tape CMCA** (3+1D architecture and geodesics): doi.org/10.5281/zenodo.20465805
- **P48 — The Complete GTE Framework** (main synthesis monograph): doi.org/10.5281/zenodo.20560550

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

[1] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl}. Zenodo, 2026. URL https://doi.org/10.5281/zenodo.20560550. https://doi.org/10.5281/zenodo.20560550. Preprint.

[2] Nova Spivack. Emergent gravity from the ϕ_{mdl} field: Einstein equations, kink sources, and quantum gravity scale. Zenodo, 2026. URL https://doi.org/10.5281/zenodo.20417559. https://doi.org/10.5281/zenodo.20417559. Paper P38 in the UGP Physics series.

[3] Nova Spivack. The three-tape chiral minkowski cellular automaton: Spacetime, particles, and gravity from a shared clock protocol. Preprint (Zenodo), 2026. URL https://doi.org/10.5281/zenodo.20465805. https://doi.org/10.5281/zenodo.20465805. UGP Physics series, P45.